

👉 XAI: Practitioner Interpretability Needs

FAMAF - UNC

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**eXplainable
Artificial Intelligence**

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

The Gap in Interpretable ML Research

- **Current Focus: (of 2023)**
 - ▶ Developing new interpretable models and explanation methods
- **Much Less Explored:**
 - ▶ How useful are these tools actually to users?
 - ▶ Do practitioners effectively use interpretability tools?

- ① **Human Factors in Model Interpretability** by Hong et al.
 - ▶ Industry practitioners' needs and uses for interpretability
- ② **Interpreting Interpretability: Understanding Data Scientists' Use of Interpretability Tools for Machine Learning** by Kaur et al.
 - ▶ How practitioners actually use interpretability tools

These papers explore:

- How do ML practitioners use interpretability tools?
- What are their unmet needs?
- Do interpretability tools actually help practitioners understand models?

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Human Factors in Model Interpretability: Industry Practices, Challenges, and Needs

SUNGSOO RAY HONG, New York University, USA

JESSICA HULLMAN, Northwestern University, USA

ENRICO BERTINI, New York University, USA

As the use of machine learning (ML) models in product development and data-driven decision-making processes became pervasive in many domains, people's focus on building a well-performing model has increasingly shifted to understanding *how* their model works. While scholarly interest in model interpretability has grown rapidly in research communities like HCI, ML, and beyond, little is known about how practitioners

Key Contributions:

- 1 Conducts interview study to understand industry practitioners' existing needs and uses for interpretability
- 2 Presents findings on roles, stages, and goals related to interpretability
- 3 Identifies aspects of interpretability under-supported by existing technical solutions



Results: Three Dimensions ---

Results Overview

Three Key Dimensions:

- ① **Interpretability Roles**
 - ▶ Who needs interpretability?
- ② **Interpretability Stages**
 - ▶ When is interpretability needed?
- ③ **Interpretability Goals**
 - ▶ Why is interpretability needed?

Interpretability Roles

Three Primary Roles:

🍵 Key Themes ---

Theme 1: Interpretability is Cooperative

Communication is Central: - Important for communicating with domain experts and stakeholders - Facilitates trust, sometimes just by virtue of including an explanation



🍷 Paper 2: Interpreting Interpretability ---

Interpreting Interpretability

Understanding Data Scientists' Use of Interpretability Tools for Machine Learning

Authors: - Harmanpreet Kaur, Harsha Nori, Samuel Jenkins (University of Michigan) - Rich Caruana, Hanna Wallach, Jennifer Wortman Vaughan (Microsoft Research)

Research Question

Large Scale Survey ---

Survey Methodology

Study Design:

- **Type:** Survey based on example queries from previous tools
- **Participants:** 197 from mailing list of large tech company
- **Analysis:**
 - ▶ Coded open-ended responses
 - ▶ Statistical tests to compare outcomes by condition

Experimental Conditions

Two Factors: Explanation Type

- GAM
- SHAP

Visualization Type

- Normal (correct)
- Manipulated (obviously wrong)

Key Question

Discussion and Implications ---

Key Takeaways from Both Papers

Paper 1: What Users Need

- Cooperative interpretability
- Process-oriented tools
- Context-dependent explanations
- Post-deployment support

Paper 2: How Tools Perform

- Over-trust is common
- Misuse is frequent
- Better tools needed
- Training is essential

Critical Questions for Discussion

① Better tools vs. better training?

- ▶ Should we focus on improving tools or educating users?

② Persuasion without understanding?

- ▶ Is it problematic if explanations convince without comprehension?

③ Accessibility vs. sophistication?

- ▶ How to make tools accessible without oversimplifying?

④ Role-specific design?

- ▶ Should we design different tools for different roles?

Thank You!

Questions and Discussion